

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 85. Data Cleaning - Project

Section 1: Lecture Summary

The lecture focuses on a practical project involving data cleaning, specifically cleaning a scanned handwritten note string which contains alphabets, spaces, and unwanted special characters or numbers due to scanning errors. The task is to write a Python program that takes a given string (simulating a scanned text) and removes all characters except alphabets and spaces, replacing any special characters or digits with spaces. The instructor demonstrates the problem context — showing how handwritten notes scanned electronically can produce strings where spaces are often misread as some random symbols. The solution uses Python string methods to iterate through characters, check if each character is alphabetic or a space, and build a clean string by including valid characters or substituting invalid ones with spaces. Finally, a simple Python program for this cleaning task is written and explained.

Section 2: Key Concepts and Explanations

- Problem Context

- Handwritten notes are scanned and converted into text by a scanner.
- Scanner errors include misreading spaces as special characters or numbers.
- Aim: To clean the scanned string by removing all characters except alphabets and spaces. Errors must be replaced by spaces.

- Data Cleaning

- In data preprocessing, cleaning means removing unwanted characters or fixing format issues to prepare data for further processing.
- This project shows a basic cleaning technique focusing on character filtering.

- String Handling in Python

- Use a variable (e.g., `scan`) to store the raw input string with errors.
- Use another variable (e.g., `clean`) initiated as an empty string to hold the output.
- Loop over each character in the scanned string.
- Use string methods:
 - `str.isalpha()` to check if the character is an alphabet letter.
 - `str.isspace()` to check if the character is a space.
- If the character is alphabet or a space, append it to the `clean` string.
- Otherwise, append a space instead (to replace special chars or digits).

- String Concatenation

- Strings are immutable in Python; concatenating inside a loop creates new strings each time, but this is acceptable here for simplicity.

- Output

- The final `clean` string contains only alphabets and spaces, suitable for further text processing or analysis.

Section 3: Example Code and Use Cases

```
scan = 'These+notes#reveal9Newton seeking-out an(!underlying structure  
to/the\\pyramid'  
  
clean = ''  
  
for x in scan:  
    if x.isalpha() or x.isspace():  
        clean += x  
    else:  
        clean += ' '  
  
print(clean)
```

Explanation:

- ``scan`` contains the noisy scanned string with special characters and numbers.
- Loop iterates over every character ``x`` in ``scan``.
- If ``x`` is a letter (``isalpha``) or whitespace (``isspace``), it is retained.
- Otherwise, replaced with a space `` ``.
- The result is printed: a cleaned string with only letters and spaces.

Section 4: Key Takeaways

- Scanned handwritten notes often have character recognition errors such as misread spaces replaced by symbols or digits.
- Data cleaning can involve removing or replacing unwanted characters to normalize text data.
- Python's string methods ``isalpha()`` and ``isspace()`` efficiently test characters for cleaning purposes.
- Iterating through a string character-by-character allows selective reconstruction of cleaned text.
- Replacing undesired characters with spaces preserves word separation for readability.
- Simple Python programs can effectively implement such cleaning tasks, laying groundwork for more complex data preprocessing projects.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Data Cleaning

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Newton was certainly not alone in believing that the ancient Egyptians had access to profound knowledge that was lost to the modern world. The searching out of ancient occult secrets was a central trope of alchemy, a subject which Newton studied deeply. These notions reveal Newton seeking out the underlying structure of the universe and the unit of measurement used by its builders. Drawing on a range of ancient and contemporary sources, especially John Greaves's *Pyrætopographia* (1646), he compared a wide range of measurements including external dimensions, the lengths of tunnels or galleries, the height of chambers, and the size of individual bricks. He aimed to prove that all of these had been calculated from a common unit of measurement: the royal cubit.

Newton hoped that quantifying the royal cubit from the Great Pyramid would in turn allow the precise dimensions of other ancient measures to be established, which would be enormously important for Newton for a number of reasons. Newton had been grappling with the theory of gravitation since the mid-1680s, but in order to demonstrate his calculations on a planetary scale he needed an accurate measure of the circumference of the Earth. Newton believed it likely that the ancients had been able to measure the Earth using techniques lost to modern man. The figure given by Eratosthenes (3rd century BCE) did not fit Newton's propositions for gravitational attraction so he turned to the earlier figure given by Thales and Anaximander in the 6th century BCE, which was that the Earth's circumference was 60,000 "stadia". Assuming that the Greeks took their measures from the Egyptians, it should therefore be possible to quantify the stadia from the cubit, and the earth from the stadia. Newton abandoned this line of argument before the publication of the *Principia*, but it is likely that when making these notes he hoped that the pyramid would give him the measure of the Earth and prove gravitational theory.

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Data Cleaning Project



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Data Cleaning Project



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Data Cleaning Project



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```
    if x.isalpha( )
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Data Cleaning Project



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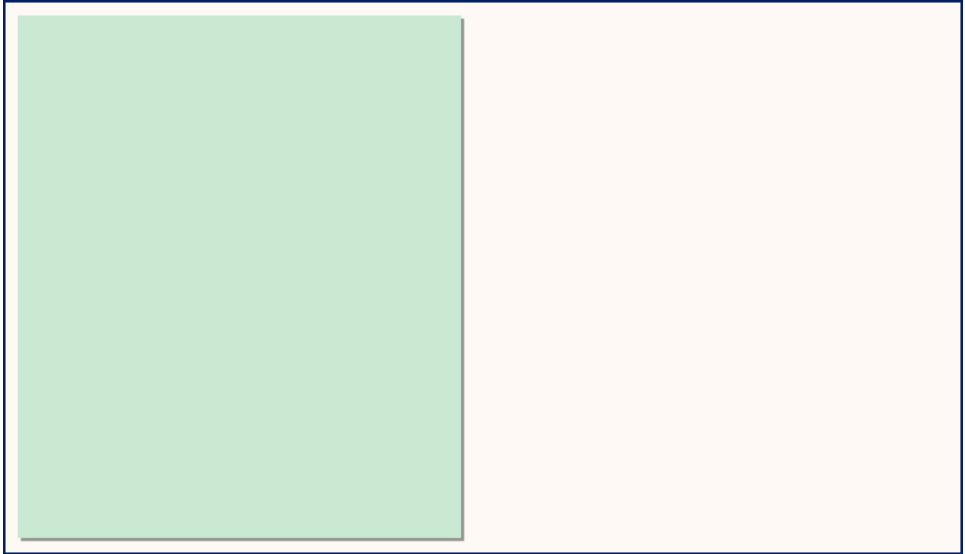
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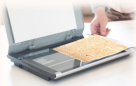


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